

Comments to the
Secretary
Federal Maritime Commission
800 N. Capital St. NW
Washington, D.C. 20573

In the Matter of:)
) FR 2011-02482
Slow Steaming- NOI)

To: The Commission

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I. Introduction

1. Pursuant to the notice of inquiry (NOI) published by the Federal Maritime Commission (“the Commission” on January 31, 2011 regarding the solicitation of views on the impact of slow steaming, I comment on my own behalf as a citizen interested in issues pertaining to maritime and commercial law. In addition to my interest in the commercial ramifications of maritime law, I am intrigued by international shipping as I have spent the majority of my life watching tanker and freight ships coming in and out of harbors having lived in Charleston, SC and Wilmington, DE, two east-coast cities that boast a large shipping presence.
2. I intend to comment on the five inquiries for interested parties. Slow steaming appears to be a tremendous solution to a cocktail of issues that confront the shipping industry. Even though more research and demonstrations must be accomplished prior to the global implementation of this practice can occur, slow steaming is destined for a useful and bright future. Slow steaming, when utilized correctly, will greatly enhance the ability of the shipping industry to thrive, especially in this uncertain time with high bunker fuel prices and increased attention and research directed to the impact of fossil fuels on the global climate.

II. Slow Steaming and Why it Needs to be Addressed

3. As the mission of the Commission is to foster a fair, efficient and reliable international ocean transportation system, it is highly commendable that the agency has issued a NOI to gather more information on the impact of slow steaming. Many ocean liner carriers that are regulated by the Commission have implemented the practice of slow steaming over the past

two years. This practice has significant economic consequences to the shipping industry as well as environmental ramifications.

4. Slow steaming is a practice of ocean liner carriers in which the normal speed of the ships is reduced. These measures were first implemented in response to the depressed international trade conditions following the economic downturn of 2008. The practice of slow steaming can also be used to reduce greenhouse gas emission to comply with environmental initiatives and concerns. Slow steaming can address these two issues at once.
5. Technically speaking, a ship that reduces speed by 20% will use 40% less fuel and reduce the corresponding CO2 emissions. The ships will travel at 20-22 knots as opposed to the standard 25 knots. In order to compensate for the slower speeds, 1-2 extra vessels are added to each route. Even with the extra ships, emissions have still shown a decrease of roughly 7% per container moved. It also means stretching out each route by one or more weeks, resulting in a reduction of number of miles travelled in a year by each vessel, which may in turn result in a longer lifespan.
6. Slow steaming is a complex issue that creates advantages and disadvantages for carriers and shippers depending on the trade conditions as well as the commodities being shipped. In order to judge the future of slow steaming, one must be able to determine the effect of adding extra vessels to accommodate for lost time as well as the impact of increases in inventory costs.

III. Five Inquiries

a. What are the major benefits and costs associated with slow steaming?

7. Slow steaming has greatly benefited the shipping industry in the face of rising bunker fuel costs and helped the shippers to weather the economic downturn of 2008. Many benefits and costs exist though in this practice.
8. The global economic crisis crippled or chilled many industries. The international shipping container industry suffered the worst year in its 45-year history from late 2008 to 2009. The U.S. liners had a rapid recovery however in 2010 as container volumes expanded by 11% and the total volume of liner imports to the U.S. increased 13% following a decline of 14% in 2009. The gains however did not erase the deficit exacerbated in part by the global economic downturn as the total container volume in U.S. trades were still 5% below their 2007 peak.
9. Slow steaming greatly helped the shipping industry stay afloat during this time. An oversupply of tonnage and a decline in demand occurred with many currencies and businesses in financial uncertainty and flux. Additionally, there was an addition of ships to the fleet, but not enough demand to accommodate the new carriers and they faced the possibility of lying idle. At one point in 2009, 575 ships, or 12% of the worldwide fleet capacity was idled. This had been largely remedied by the end of 2010 as the worldwide idle capacity was only 2%. Thus, the total freight revenue grew and the companies' were also able to expand their service offerings. Slow steaming not only remedied this situation by putting more ships into use, but also by reducing fuel costs even though more ships were active.

Additionally, the extra ships in use allowed the companies to balance the international supply and demand.

10. Certain situations lend themselves to slow steaming while others do not. For example, the practice will be attractive when there are higher bunker fuel costs and low charter rates, but a lower fuel cost could result in a different situation. Additionally, carriers of certain types of goods, such as meat, produce or high-value commodities may not see the financial benefits of slow steaming due to spoiling, less shelf-time at grocery or an added delay in payment, respectively.
11. One shipping executive described the practice as a “win-win-win” for their customers, the environment and their business. The money saved with slow steaming can be reinvested into the corresponding corporation leading to further innovation and efficiency throughout the shipping process.
12. Although there is much analysis as to the benefits of the carriers (as noted above), there is little information about how slow steaming affects the seller and consumer. One shipping executive noted the importance of customer satisfaction in that, “We believe we serve our customers best by steadily improving schedule reliability, by keeping fuel costs down, and by continuing to improve on our carbon footprint.”¹ Although some customers have complained of longer inventory times, analysis has helped to show that slow steaming helps prevents bottlenecks on terminals.
13. It is unclear whether the cost savings from slow steaming have benefited the customers, either in discounted or adjusted rates. As shippers work for the customer, it is vital that the customers realize these savings. Similar to a stock dividend, certain customers will prefer instant financial gain in discounted travel rates and others will prefer that the shippers reinvest the savings in order to enhance service through innovation in the future. Granted, the shippers are a for-profit entity and are entitled to cutting costs. However, from an economic point of view, competition will eventually drive the customers towards the shippers who do grant them some financial benefit from the slow steaming.
 - b. To what extent has the slow steaming of services in the U.S. ocean liner trades reduced greenhouse gas emissions?
14. Studies by both ocean liner companies as well as independent organizations have shown, on average, an 11% reduction in CO2 greenhouse gas emissions since 2008 due to slow steaming. These studies used complex formulas to truly analyze and understand the impact of slow steaming from a variety of angles in order to better improve the practice and industry. These studies took into account specific shipping routes as opposed to the global industry as certain paths utilize slow steaming more than others, and especially Pacific and European trades. There is a global industry initiative to reduce greenhouse gas emissions 15% by 2018, and at its current rate is on path to reach that number in the very near future.

¹ Slow Steaming Here to Stay, MAERSK, 9/1/2010, <http://www.maersk.com/AboutMaersk/News/Pages/20100901-145240.aspx>.

- c. Discuss the likely long-term prevalence of slow steaming and its potential impacts on the economy and/or the environment.
15. This practice also has environmental concerns. Since container shipping (roughly 4% of the fleet) accounts for nearly 20% of CO2 emissions from shipping, it is crucial to address a solution that can lower this number.
 16. Slow steaming saves energy and cuts CO2 emissions in addition to taking boxes out of circulation, which helped shipping rates from falling even more. Nearly 4.1% of the global fleet was slow steaming, a process that helped to balance out supply and demand.
 17. Schedule reliability has improved as well due to the fact that the slower speed allows the vessels to adjust speed continuously to deliver exactly on time. Slow steaming did not take off at first, but has helped Maersk cut CO2 emissions per container by 12.5% from 2007 to 2009.
 18. The turmoil in the Middle East is placing another hurdle in the recovery of the international shipping industry. Oil prices have been steadily rising since the uprisings and demonstrations began a few months ago due to, among other things, instability in the region. At the moment, oil prices are hovering, and have been for quite some time, above the \$100-per-barrel line. With no certain end in sight for the revolutions and unrest in the world's most important oil producing region, higher bunker fuel costs could be an everlasting feature, furthering the practice and need for slow steaming.
 19. The global demand may not be as affected if the prices stay below \$120/barrel, but "if it goes up to US\$200-250, then the demand equation will look different. You're going to probably see some dampening of demand, particularly in the US and Europe markets which are more sensitive."² Additionally, oil is a fossil fuel. Our dependence on it is well documented and many energy companies are in the process of creating new efficient and economical fuel sources. Until that point is reached however we will still be dependent on the oil, especially for these large ships. Slow steaming will deplete these fuel reserves at a slower rate than before, therefore allowing more time to find a replacement.
 20. The rising oil prices have increased bunker fuel costs and the Bunker-world Index has already climbed from 1,200 to 1,435 this year. The brunt of this increase will be felt by the shippers, who must deal with bunker surcharges or an increasing bunker adjustment factor (BAF). These effects will be felt down the line beginning in April as the BAF levels are adjusted on a monthly basis
 21. Once again, slow steaming appears to be the answer to the high bunker fuel costs. One shipping spokesperson stated, "we expect bunker prices to remain high for the rest of the year. As a result, we foresee more slow-steaming by carriers to offset the cost increase."³ Although slow steaming is a possibility, the bunker price increases have outstripped the cost

2 Joyce Hooi, Bunker Prices Rising Fast on Oil Price Hikes, Business Times (Singapore), March 2, 2011, Top Stories Section.

3 Id.

savings of slow steaming. For example, Maersk cut bunker consumption by 10% per 40 foot equivalent unit by slow steaming, yet bunker costs rose 26%.

22. Shipping corporations have seen success in the implementation of slow steaming. Maersk, the largest shipper in the world has seen fuel consumption reductions of up to 30% on certain routes. These savings help to mitigate the cost of having crews at sea for an extended period of time, possible engine damage and longer delivery schedules.⁴ Another major player in the shipping industry, China COSCO, implemented super slow steaming after the initial hikes in fuel prices following the economic downturn. Super slow steaming involves reducing the speed of the ships to 14 knots on certain routes and the group has reduced the overall speed of its ships by an average of 12%. COSCO saved as much as \$1 million annually due to the reduced speeds in addition to cutting CO2 emissions by roughly 10,000 tons.⁵ Another carrier, Orient Overseas Container Line, cut fuel consumption by 7% as a result of slow steaming in the first half of 2010.⁶
23. The increased usage of slow steaming, and extra ships, will help to absorb some of the oversupply of tonnage potentially available.⁷ One expert has estimated that slow steaming has saved the members of the Transpacific Stabilization Agreement, a group that carries about 95% of seaborne cargo across the Pacific Ocean, around \$2 billion over the past 18 months.⁸ Carriers have been able to offset the higher fuel costs through a fuel surcharge levied using a special formula on cargo shipped by exporters, although this formula has not seen an adjustment to reflect the savings from slow steaming.⁹
24. One potential issue that may affect the viability of slow steaming, or super slow steaming in particular, is that certain goods may not be appropriate due to their nature for the practice. One of these goods is produce and chilled meats, a product vital to New Zealand's trade. Since the trips will be routinely at least two to six days longer due to slow steaming, the perishable items will have a shorter shelf life at grocery stores and their value will be reduced. Although the chairman of the New Zealand Shippers Council does not believe slow steaming to be a necessarily urgent issue, he does address the concern regarding slower speeds or super slow steaming. He claims that "those [perishables] export industries would not exist. They would not be able to export to destinations like Europe if carriers reduced speed any more from New Zealand."¹⁰ The longer journey will reduce the price of the goods due to a shorter shelf life and could possibly leave it with no value at all if the ship arrives at the wrong time. A meat industry executive has claimed "super slow steaming could put an end to selling chilled meat in Europe, by cutting the shelf life even further." The chilled-lamb industry exports to Europe are worth about \$700 billion, which is almost entirely sent

4 Hoover's In-Depth Company Records, Maersk Inc., March 9, 2011.

5 Hoover's In-Depth Company Records, China COSCO Holdings Company Limited, March 9, 2011.

6 Keith Wallas, Shipping Lines Asked to Report on Benefits of Slow Steaming, South China Morning Post, Feb. 7, 2011, p. 1.

7 Joyce Hooi, Dry Bulk Sector Choppy by Containers Calm: Aussie cyclones add to dry bulk woes but new supply won't flood containers, Shipping Times (Singapore), Feb. 24, 2011, Shipping News Section.

8 Wallas at. 1.

9 Id.

10 James Weir, Slow-boat Shippers Have Exporters Steaming, The Marlborough Express, Jan. 12, 2011, p. 14.

by sea. Prices and demand are currently at an all-time high for this premium product but it may be snuffed out by the implementation of even slower speeds.

d. How important is slow steaming in the overall effort to reduce emissions of greenhouse gases and other air pollutants arising from ocean liner operations?

25. Slow steaming is presently vital to overall effort to reduce emissions and other air pollutants from ocean liner operations. To begin with, it has been proven to expend less fuel than if the ships were moving at normal operating speeds. This will save fuel, delaying the inevitable total depletion of the world's fuel inventory. Additionally, the less fuel that is used directly correlates to a decrease in the amount of greenhouse gases emitted from the ocean liners. It is a win-win as less fuel will be used overall as well as a large decrease in the amount of emitted greenhouse gases, such as CO₂ and SO₂.

26. Most importantly, slow steaming is performed without any major new technological innovations or improvements to the design of the ship or the engine. Nearly any ship is presently capable of slow steaming, without the need for a costly upgrade or the purchase of expensive new equipment. This benefits many as well as helps to facilitate the spread of the practice since the industry will not have to spend any money in order to save, an act that would most definitely slow up the process as well as the efficiency of shipping.

27. Newer technological innovations may, and most likely will, bring about more efficient and economical solutions in the future. Until then though, the shipping industry can act proactively to help solve a global crisis in terms of greenhouse gases and additionally respond to the recovering economy and fuel market in an economical fashion.

e. What data sources are available to measure the economic and environmental impacts of slow steaming?

28. Although it is a relatively new practice, certain institutions and corporations have stepped forward and researched the economic and environmental impacts of slow steaming. Certain shippers such as Maersk and China COSCO have demonstrated through their own implementation and research that slow steaming does save fuel, cut down greenhouse gas emissions and save money. Slow steaming has not only created a solution to the global fuel crisis, but also to the idled ships and oversupply of tonnage worldwide.

29. Other institutions such as the International Shipping Organization, Elsevier and the Review of Maritime Transport have performed research and analysis regarding the economic and environmental impacts of slow steaming. Their published results paint a positive picture of slow steaming and its future. In addition to creating metrics with which to study and interpret the findings, these organizations have also helped to give credence to the practice by demonstrating that slow steaming does in fact help make economic and environmental sense.

30. One study in particular, Elsevier, dove into the analysis of slow steaming with an array of formulas and charts that depict the realities of the practice visually.

31. This study used secondary data to analyze the impact of slow steaming on CO₂ emissions since 2008. The study did not deal with the global scale but instead specific trades that were subject to different rates of slow steaming.¹¹ A formula was created for a containership with a capacity of more than 1000 TEU using a two-stroke marine diesel engine as follows. The formula can prove incredibly helpful in the Commission's decision to study the effects of slow steaming and is republished and described below:

1. Impact of slow steaming on CO₂ emissions

$$\Delta CO_{2,ds \rightarrow ss} = 3.17 \times \sum_{k=1}^n (ME_{k,sea} \times D_{k,sea} + ME_{k,port} \times D_{k,port}) = 3.17 \times \Delta FC_{ds \rightarrow ss}$$

$$ME_{k,sea} = SFOC_k \times EL_k \times kWh_k$$

32. Reducing the speed from design speed (V_{ds}) to slow steaming (V_{ss}) will impact the main engine fuel consumption at sea (ME_{k,sea}) and will have a smaller impact on the auxiliary engine.¹² D_{k,sea} is the number of days at sea and D_{k,port} is the number of days in ports. The main engine's daily fuel consumption at sea results from the specific fuel oil consumption (SFOC_k), engine load (EL_k) and engine power (kW_h). The ships are built for moving at close to design speed and SFOC is optimal at this level. For the study, it was assumed that fuel consumption in port was 5% of the main engine consumption at design speed.¹³ Using this formula, it was found that the impact of slow steaming will depend on the speed reduction as long as it is reduced up to 10-15%.¹⁴ This study required information on the initial vessel's fuel consumption at sea at design speed (ME_{k,sea}) and service characteristics such as number of vessels deployed and slow steaming, as well as days at port and sea. The study gathered information from Lloyd's Register-Fairplay for the daily fuel consumption for 451 container vessels.¹⁵ Additional factors were used and can be seen in the charts found in their published study.
33. One of the biggest findings in these studies was that the long-term sustainability of slow steaming depends on the additional operational costs for the vessels that are necessary and the changes in the inventory costs.¹⁶ Additionally, as long as the current bunker price is higher than the break-even bunker point (which is determined in the formula below and described in the study), slow steaming is an attractive option and the CO₂ emissions reductions will continue.¹⁷

¹¹ Pierre Cariou, Is Slow Steaming a Sustainable Means of Reducing CO₂ Emissions from Container Shipping?, ElSevier, Transportation Research Part D: Transport and Environment, Volume 16, Issue 3, 2011 pp. 260-264.

¹² Id.

¹³ Id. at 261

¹⁴ Id.

¹⁵ Id.

¹⁶ Id.

¹⁷ Cariou at 261.

2. Long-term sustainability of slow steaming

$$BP^* \geq \frac{OC_{\Delta n, ds \rightarrow ss} + \Delta Rot_{ds \rightarrow ss} \times IC_{teu}}{\Delta FC_{ds \rightarrow ss}}$$

34. Basically, the break-even point for which the reduction is sustainable occurs when the bunker price is greater than the change in operational costs and miles traveled divided by the change in fuel costs. This study did not go into a variety of technical elements but noted that at very slow speeds the quality of exhaust is altered and these slower speeds can lead to design and safety issues.¹⁸ In conclusion, the study demonstrated the slow steaming has reduced emissions by 11% since 2008 and was done without the adoption of any new technology. The practice is fragile in the long run as the volatility of bunker prices and cost of inventory is always a factor in prices and practices.
35. There are many sources of analysis and study for a practice that is so young. The continued application and use of slow steaming will lend itself to more study so that a comprehensive answer can be attained regarding its implementation.

IV. Conclusion

36. Slow steaming is an interesting innovation that has successfully lowered operating costs and CO2 emissions without any adoption of new technology. Additionally, it is a practice that has helped carriers reduce the need to lay off ships. Although there are certain situations in which slow steaming may not be as useful, on the whole it is an exciting new industry practice that can help to combat the growing global fuel issue, economic crisis and worldwide environmental situation at the present while the money saved can be reinvested with the potential of new innovations and savings to the customer.

¹⁸ Id.